Biochemistry Cumulative Exam
March 6, 2009

Cytochrome b_5 oxidase is an integral membrane protein in the mitochondrial respiratory chain of a newly discovered yeast which transfers four electrons from four molecules of cytochrome b_5 to one molecule of O_2 to form two molecules of water, and pumps 6 protons from the inside to the outside of the mitochondrial membrane. Cytochrome b_5 has a redox potential of $E_o' = +0.02$ V, and O_2 has a redox potential of $E_o' = +0.82$ V. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.2$ V, and the pH inside and outside the respiratory membrane was $pH_{in} = 8$, and $pH_{out} = 6$.

1. (10) Give a simple diagram showing how cytochrome b_5 oxidase couples the energy of electron transfer to pumping protons across the mitochondrial membrane.

2. (25) Calculate the free energy change ΔG^o for electron transfer from 4 molecules of cytochrome b_5 to one molecule of O_2 to form two H_2O. (not including the energy of proton pumping).

3. (40) Give a formula for the free energy change ΔG for pumping protons from the inside to the outside of the mitochondrial membrane, and calculate ΔG for the proton pumping component of the cytochrome b_5 oxidase reaction under the above conditions.

4. (25) Calculate how efficiently cytochrome b_5 oxidase couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping under the above conditions.

Possibly useful formulas:

$\Delta G^o = - RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$; $\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

Membrane Transport: $\Delta G(X_{in} \rightarrow X_{out}) = RT\ln([X_{out}]/[X_{in}]) + ZF(V_{out} - V_{in})$.

Z = charge on X

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol